ENGINEERING DOCUMENT QA

Run 1 vs. Run 2 — Baseline and Improved-Instructions Comparison

Document: DEMO-PMP-001 | **Equipment:** Fictional centrifugal pump P-101

Purpose. Compare the two ChatGPT-assisted review outputs to see whether more explicit instructions produced a more structured and auditable documentation-gap assessment.

1. Summary of observed differences

Measure	Run 1 — Fresh baseline	Run 2 — Improved instructions
Findings register	9 findings: 6 High, 3 Medium (per Run 1 report).	13 findings: 10 High, 3 Medium (per Run 2 report).
Granularity	Several related issues are grouped together.	More distinct items, including safe-opening conditions, part verification, pre-start checks, restart authorization, and post-start monitoring.
Priority rationale	Preliminary priorities are given.	Adds an explicit rationale for each finding and explains what confidence means.
Confidence	No per-finding confidence rating.	Adds per-finding confidence ratings, explicitly limited to the gap in the supplied excerpt.
Verification planning	Verification sources are listed for findings.	Adds a consolidated verification plan with evidence expected to close each group of findings.
Checklist	More detailed checklist of missing items and unverifiable external facts.	Uses three explicit statuses: addressed, missing, or unverifiable; reasons are provided.

2. What improved in Run 2

• **More granular coverage:** Run 2 separates safe-opening prerequisites from isolation verification and separately addresses replacement-part suitability, pre-start readiness, restart authorization, and post-start monitoring.

• **Better auditability:** Each finding includes source-step evidence, a stated gap and impact, preliminary priority rationale, corrective action, verification source, and confidence.

• **Clearer closure path:** The verification plan describes what evidence should demonstrate resolution rather than merely listing missing information.

• **More careful uncertainty handling:** The report states that a gap in the supplied excerpt does not prove the control is absent from the wider site system.

3. Interpretation and limitations

The observed improvement is in the structure and granularity of the generated review, not proof that all findings are technically correct or that an autonomous agent improved itself. Finding counts are not a standalone quality metric: additional findings can represent useful separation, but may also create overlap. A qualified reviewer should check whether the 13 Run 2 findings are distinct and proportionate.

Both reviews are based only on a fictional six-step procedure. No OEM manual, site procedure, engineering drawing, inspection record, physical inspection, or formal risk assessment was used. Priorities are preliminary documentation-review judgments, not formal risk ratings.

4. Execution transparency

These two reports were generated in separate ChatGPT conversations using different instructions. This comparison records the outputs available for review. No external agent platform, independent autonomous execution, or separate tool trajectory has been demonstrated in this project. The work should therefore be described as a ChatGPT-assisted iterative document-review case study unless additional agent execution evidence is collected.

5. Recommended next step

Retain both source reports and this comparison. Update the repository README to explain the prompt change, summarize the observed output differences, and state the limitations. Do not claim validation against real pump requirements or autonomous agent deployment.

Training exercise only — not an operational maintenance procedure or approval.